



What's a prompt?

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1. Definition

A **prompt** is the textual (or multimodal) input you feed to a Large Language Model (LLM) so it can generate an output. Think of it as **the instruction set + context** that frames the model's task. The quality of the prompt largely dictates the quality of the response.

2. Why Prompts Matter

- **Steering:** Prompts tell the model *what* to do and *how* to do it.
- **Boundary-setting:** They define length, style, tone, and format constraints.
- **Disambiguation:** A precise prompt reduces vagueness, lowering the risk of irrelevant or hallucinated content.

3. Anatomy of a Strong Prompt

Component	Purpose	Example Snippet
Role / Persona	Aligns voice & expertise.	"You are a cybersecurity analyst..."
Task Instruction	States the main goal.	"...summarize the breach report..."
Context / Data	Supplies necessary facts.	"...using the log excerpts below."
Constraints	Sets boundaries (length, format, style).	"Limit to 200 words, bullet points only."
Examples (Few-shot)	Demonstrates desired patterns.	"Input → Output" pairs.

Rule of thumb: Include only the context required for success; extra noise can dilute performance.

4. Core Prompt-Design Principles

1. **Clarity** – Use unambiguous language (“Summarize” vs. “Tell me about”).
2. **Specificity** – State concrete requirements (word limits, audience level, tone).
3. **Structure** – Organize instructions in logical order or with headings.
4. **Goal Orientation** – Tie every instruction to the final deliverable.
5. **Incremental Iteration** – Refine prompts based on observed outputs.

5. Prompting Strategies

Strategy	When to Use	Quick Example
Zero-shot	Straightforward tasks; no examples needed.	“Translate this sentence to Japanese.”
Few-shot	When output style matters.	Provide 2–3 Q&A pairs before the new question.
Chain-of-Thought	Complex reasoning/math.	“Let’s reason step by step...”
System/User/Assistant Roles	Long dialogs or multiple constraints.	System: rules; User: request; Assistant: response.
Self-Critique / Refinement	Quality assurance.	“Provide the answer, then critique and improve it.”

6. Good vs. Poor Prompt Examples

Poor:

“Explain climate change.”

Improved:

“You are an environmental journalist. Write a 3-paragraph explainer on climate change for high-school students (age 14-18). Include one real-world example and end with a call-to-action. Target length: 250-300 words.”

7. Common Pitfalls

- **Under-specification:** Vague asks produce vague answers.
 - **Overloading:** Cramming multiple unrelated tasks in one prompt.
 - **Hidden Assumptions:** Forgetting to spell out domain-specific nuances.
 - **Moving Targets:** Changing requirements mid-conversation without resetting context.
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8. Quick Prompt-Building Checklist

- ☐ Define the **role** or persona you want the model to adopt.
 - ☐ State the **primary task** in plain language.
 - ☐ Provide **relevant context or data**—nothing extraneous.
 - ☐ Specify **format, length, tone, or style** constraints.
 - ☐ (Optional) Add **few-shot examples** if style consistency is critical.
 - ☐ Review for clarity; trim ambiguity.
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9. Key Takeaways

- A prompt is the **interface layer** between human intent and model behavior.
- **Clear, specific, structured, goal-driven prompts** reliably yield accurate, relevant outputs.
- Iterate and refine; prompt engineering is an **experimental discipline**—observe, tweak, repeat.

Bottom line: Mastering prompt design equips you to harness the full power of LLMs, transforming raw model potential into actionable, high-quality results.